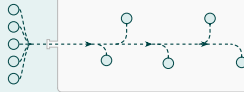


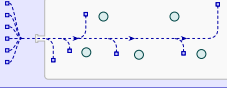
Initialize Run: homogeneous genotype A_0
with $B_0=1-A_0$; start population $N_0=100$;
fix sampled parameters



For each generation – $g = 1, \dots, 1,000$

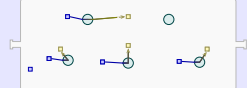
1. Resource Inflow

Add fresh M_1 to the well-mixed environment pool



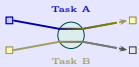
2. Task A

Convert $M_1 \rightarrow M_2$; task done by organisms according to their A_i



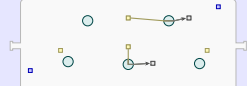
4. Energy Accounting

Credit Task A and Task B yields to $E_i^{(g-1)}$, then deduct maintenance c_m :
 $E_i^{(g)} \leftarrow E_i^{(g-1)} + A_i + Y \cdot B_i - c_m$



3. Task B

Convert $M_2 \rightarrow$ waste; task done by organisms according to their B_i



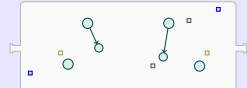
5. Death

Draw $P_{\text{death},i}(E_i)$ for each organism; remove dead organisms



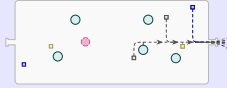
6. Duplication

Draw $P_{\text{dup},i}(E_i)$ for each organism; split energy between parent and offspring; offspring receives a copy of the parent's traits (A_i, B_i)



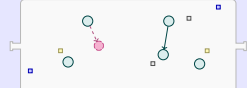
8. Chemostat Flow

Draw $P(\text{outflow}_i)$ for each survivor; remove organisms and scale environmental M_1 & M_2 by $(1 - \phi)$



7. Trait Mutation

With probability μ_m , modify offspring traits within $\pm\sigma_m$ of the parental value; see Supplementary Methods



$g < 1,000$,
 $N_g > 0$

N_g – final population size at generation g

9. End of Generation

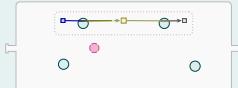


$g = 1,000$,
 $N_g > 0$

$N_g = 0$

10. Cross-Feeding Outcome Analysis

Evaluate final population for Persistence, Specialization, and Exchange (Methods); record hit or non-hit for this parameter draw



Run Ends (Collapsed)

Population extinct before generation 1,000; no outcome scoring

No Organisms Left

$\blacksquare$ M_1 $\blacksquare$ M_2 $\square$ Waste $\circ$ Organism

$\bullet$ Mutant offspring